

Minwoo Kang, Ph.D.

Postdoctoral Scholar · HR Thiam Lab · Department of Bioengineering, Stanford University
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RESEARCH INTERESTS

Cellular and nuclear biophysics, Mechanobiology, Cell mechanics, Quantitative biology, Live-cell microscopy, Biomedical image processing, Atomic force microscopy (AFM), Neutrophils, NETosis, Fibroblasts, Tumor microenvironment (TME), Wound healing process, Scar formation, Cancer-associated fibroblasts (CAFs), Epithelial-mesenchymal transition (EMT), AI-based cellular phenotyping

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Feb. 2018 – Feb. 2023
Department of Mechanical Engineering — Ph.D. Daejeon, Republic of Korea
Advisor: Prof. Jennifer H. Shin (Department of Mechanical Engineering, KAIST)
Dissertation: Mechanical Perspectives on Fibroblasts in Wound Healing and Tumorigenesis

Soongsil University Mar. 2015 – Feb. 2017
Department of Mechanical Engineering — M.S. Seoul, Republic of Korea
Advisor: Prof. Sang-Ho Suh (Department of Mechanical Engineering, Soongsil University)
Thesis: Study on Geometry Improvement of Francis Turbine Runner to Reduce the Sediment Erosion

Soongsil University Mar. 2009 – Feb. 2015
Department of Mechanical Engineering — B.S. Seoul, Republic of Korea
Advisor: Prof. Sang-Ho Suh (Department of Mechanical Engineering, Soongsil University)
Thesis: Study of Valid Inlet Boundary Condition of Upper Airway Analysis considering Respiration

HONORS AND AWARDS

CRI Irvington Postdoctoral Fellowship 2026
Cancer Research Institute (CRI)

Stanford School of Medicine Dean's Postdoctoral Fellowship 2025
Stanford University

Best Paper Award 2023
Korean Society for Precision Engineering

Venture Research Program for Master's and PhD Students in the College of Engineering 2019
KAIST

Outstanding Poster Award 2017
10th International Conference on Computational Heat, Mass and Momentum Transfer

Best Presentation Award 2016
9th National Congress on Fluids Engineering

Gold Prize 2014
4th Soongsil Engineering Award

Excellence Award (3rd Prize) 2014
National College Student Fluid Engineering Competition, Annual Meeting of Fluid Engineering Division, KSME

Excellence Award (3rd Prize) 2014
4th Capstone Design Award, Soongsil University

RESEARCH EXPERIENCE

Postdoctoral Scholar, Department of Bioengineering, Stanford University

Stanford, CA, U.S. · Oct. 2023 – Present

Advisor: Prof. Hawa Racine Thiam (Departments of Bioengineering, Microbiology & Immunology, Stanford University; Sarafan ChEM-H Institute Scholar; Esther Ehrman Lazard Faculty Scholar; Chan Zuckerberg Biohub Investigator)

- **Biophysical mechanism of nuclear envelope rupture during NETosis**
 - Measuring nuclear envelope rupture force and intranuclear pressure of neutrophils undergoing NETosis with AFM
 - Mathematical modeling of nuclear envelope rupture

Guest Researcher, National Institute of Biomedical Imaging and Bioengineering (NIBIB), NIH

Bethesda, MD, U.S. · Jul. 2024

Advisor: Alexander Cartagena-Rivera, Ph.D. (Earl Stadtman Investigator, NIBIB, NIH)

Graduate Research Assistant, Department of Mechanical Engineering, KAIST

Daejeon, Republic of Korea · Feb. 2018 – Feb. 2023

- **Cell Mechanics based Analysis of Scar Formation for Strategic Scar Treatment (\$389k, collaborative research grant with 1 collaborator, Funded by Korean Government, NRF-2017R1A2B2007673, 2017–2020)**
 - Performed the project in a leading role
 - Developed the tensile stimulation device (Domestic patent)
 - Performed the tensile stimulation experiments and analyzed the data
 - Analyzed the RNA-seq data
 - Wrote the annual reports
- **AI-based classification of phenotype and function of cancer-associated fibroblasts for novel therapeutic strategy in pancreatic cancer (\$1M, individual research grant, Funded by Korean Government, NRF-2021R1A2C3008408, 2021–2025)**
 - Conceived the project idea and wrote the grant proposal
 - Performed the project in a leading role
 - Developed the deep learning code for single-cell segmentation and biophysical feature extraction with time-series data
 - Performed the live-cell imaging and classified CAF subtypes with biophysical features using machine learning algorithms
 - Wrote the annual reports
- **Development of 3D biopatch-applied breast implant technology with anti-stress micro patterns and drug delivery system for the prevention of capsular contracture (\$1M, collaborative research grant with 3 collaborators, Funded by Korean Government, 22B0401L1, 2022–2025)**
 - Supported writing the grant proposal
 - Classified subtypes of fibroblast from different breast regions with biophysical features using deep learning and machine learning algorithms

PUBLICATIONS

Journal Articles & Preprints

1. **Kang, M.**, Cabral, A. T., Sawant, M., & Thiam, H. R. (2026). Benchmarking three simple DNA staining-based image metrics for live-cell tracking of chromatin organization. *bioRxiv*. <https://doi.org/10.64898/2026.03.30.715467> (under revision)
2. Cabral, A. T., Sawant, M., **Kang, M.**, Liangqi, X., & Thiam, H. R. (2025). Chromatin is a long-range force generator that regulates plasma membrane tension and cell integrity independently of gene expression. *bioRxiv*. <https://doi.org/10.1101/2025.10.02.680155> (under revision)
3. **Kang, M.**, Devsrasou, S., Kwon, T. Y., & Shin, J. H. (2025). EMT induction in normal breast epithelial cells by COX2-expressing fibroblasts. *Cell Communication and Signaling*, 23(1). <https://doi.org/10.1186/s12964-025-02227-7>
4. **Kang, M.**, Min, C., Devsrasou, S., & Shin, J. H. (2025). AI-driven classification of cancer-associated fibroblasts using morphodynamic and motile features. *APL Bioengineering*. <https://doi.org/10.1063/5.0250502> (selected as a featured article)
5. **Kang, M.**, Ko, U. H., Oh, E. J., Kim, H. M., Chung, H. Y., & Shin, J. H. (2025). Tension-sensitive HOX gene expression in fibroblasts for differential scar formation. *Journal of Translational Medicine*, 23(1), 168. <https://doi.org/10.1186/s12967-025-06191-1>
6. Devarasou, S., **Kang, M.**, & Shin, J. H. (2024). Biophysical perspectives to understanding cancer-associated fibroblasts. *APL Bioengineering*, 8(2). <https://pubs.aip.org/aip/apb/article/8/2/021507/3296522>
7. Devarasou, S., **Kang, M.**, Kwon, T. Y., Cho, Y., & Shin, J. H. (2023). Fibrous matrix architecture-dependent activation of fibroblasts with a cancer-associated fibroblast-like phenotype. *ACS Biomaterials Science & Engineering*, 9(1), 280–291. <https://doi.org/10.1021/acsbiomaterials.2c00694>
8. Lee, J. S., Cho, H. G., Lee, J. W., Oh, E. J., Kim, H. M., Ko, U. H., **Kang, M.**, Shin, J. H., & Chung, H. Y. (2023). Influence of transforming growth factors beta 1 and beta 3 in the scar formation process. *Journal of Craniofacial Surgery*, 34(3), 904–909.
9. Park, N., Kim, K., **Kang, M.**, Suh, S.-H., Ryu, G., Kim, S. J., & Park, T. (2017). A comparative study on the Francis turbine performances according to the different vane shape. *The KSFM Journal of Fluid Machinery*, 20(4), 19–24.
10. Kim, H.-H., Jung, Y. H., **Kang, M.**, & Suh, S.-H. (2017). A comparative study on the operation capabilities according to the different capsule shapes in pneumatic capsule pipeline. *The KSFM Journal of Fluid Machinery*, 20(4), 56–61.
11. **Kang, M.**, Park, N., & Suh, S.-H. (2016). Numerical study on sediment erosion of Francis turbine with different operating conditions and sediment inflow rates. *Procedia Engineering*, 157, 457–464.

Selected International Conference Presentations

- **Kang, M.**, Cartagena-Rivera, A., & Thiam, H. R. “PAD4-dependent chromatin decompaction and nuclear envelope remodeling drive explosive nuclear rupture during NETosis.” *Biophysical Society*, Feb. 2026.
- **Kang, M.**, Devsrasou, S., Min, C., & Shin, J. H. “Morphodynamic feature-based AI framework for classification for subtypes of cancer-associated fibroblasts.” *International Conference on Biological Physics*, Aug. 2023.
- **Kang, M.**, Devsrasou, S., Min, C., & Shin, J. H. “Classification of cancer-associated fibroblasts with morphodynamics and motility feature-based AI platform.” *International Conference on Precision Engineering and Sustainable Manufacturing*, Jul. 2023.
- **Kang, M.**, Devsrasou, S., Min, C., & Shin, J. H. “Morphodynamic and motility feature-based deep learning classification for subtypes of cancer-associated fibroblasts.” *Biophysical Society*, Feb. 2023.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Tension-sensitive HOX gene regulates scar formation.” *World Congress of Biomechanics*, Jul. 2022.

- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “HOX-driven mechanobiology of scar formation.” *IUPESM WC*, Jun. 2022.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Loss of tension in skin wound regulates scar formation.” *International Conference of the Korean Society for Molecular and Cellular Biology*, Nov. 2021.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Loss of tensile stress in skin tissue induces scar formation.” *Biomedical Engineering Society*, Oct. 2021.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Loss of tensile stress in skin wound regulates scar formation.” *Invited talk for Korea Satellite Meeting — Materials, Mimics, and Microfluidics: Engineering Tools for Mechanobiology (MBI 3M)*, Jul. 2021.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Effects of tensile stress on scar formation.” *Biomedical Engineering Society*, Oct. 2019.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Differential gene expression analysis on hypertrophic scar fibroblasts and keloid fibroblasts.” *International Conference of the Korean Society for Molecular and Cellular Biology*, Sept. 2019.

Selected Domestic Conference Presentations

- **Kang, M.**, Devsrasou, S., Min, C., & Shin, J. H. “Morphodynamics and motility feature-based AI classification for identifying subtypes of cancer-associated fibroblasts.” *Korean Society for Precision Engineering*, May 2023.
- **Kang, M.**, Devsrasou, S., Min, C., & Shin, J. H. “Identifying subtypes of cancer-associated fibroblasts through biophysical feature-based AI classification.” *The Korean Society of Mechanical Engineers*, Apr. 2023.
- **Kang, M.**, Devsrasou, S., Kwon, T. Y., & Shin, J. H. “Visualization of state change of breast epithelial cells via paracrine signaling of activated fibroblasts.” *The Korean Society of Visualization*, Dec. 2022.
- **Kang, M.**, Devsrasou, S., Kwon, T. Y., & Shin, J. H. “Fibroblast aggregates induce the epithelial-mesenchymal transition of normal breast epithelial cells in a paracrine manner.” *The Korean Society of Mechanical Engineers*, May 2022.
- **Kang, M.**, Devsrasou, S., Kwon, T. Y., & Shin, J. H. “Fibroblast spheroids induce the epithelial-mesenchymal transition of breast epithelial cells in a paracrine manner.” *Korean Society for Precision Engineering*, Nov. 2021.
- **Kang, M.**, Ko, U. H., Kim, H. M., Lee, J. W., Chung, H. Y., & Shin, J. H. “Loss of tensile stress in skin wound regulates scar formation.” *The Korean Society of Mechanical Engineers*, May 2021.
- **Kang, M.**, Devsrasou, S., & Shin, J. H. “High COX2 expression from nemotoc fibroblasts promotes YAP translocation and induces α -SMA+/FAP- CAF differentiation in vitro.” *The Korean Society of Medical and Biological Engineering*, Nov. 2020.
- **Kang, M.**, Ko, U. H., Chung, H. Y., & Shin, J. H. “Role of morphostasis regulation through tensile force on scar formation.” *The Korean Society of Mechanical Engineers*, May 2019.
- **Kang, M.**, Ko, U. H., Jeong, H. T., Oh, E. J., Kim, H. M., Kim, T. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Different homeobox gene expression in hypertrophic scars and keloids.” *The Korean Society for Biomaterials*, Oct. 2018.

Patents

1. Jennifer H. Shin, Chanhong Min, **Minwoo Kang**, Hyun Tae Jeong, Tae Yoon Kwon, & Myung Chul Kim. “A method and apparatus for classification of subtypes of cells with morphological and motility features using hybrid learning.” Jan. 2025 (Reg. No. 1027625420000, Republic of Korea).
2. Jennifer H. Shin, **Minwoo Kang**, Ung Hyun Ko, & Hyun Tae Jeong. “Multi-cell stimulation device.” Dec. 2021 (Reg. No. 1023410530000, Republic of Korea).

MENTORING EXPERIENCE

Mentored a junior lab member to win a student research grant — “Deep learning-based quantitative evaluation indicators for the neural differentiation,” *Venture Research Program for Master's and PhD Students in the College of Engineering* (May 2022 – Dec. 2022) May 2022

Mentored a junior lab member to win a student research grant — “AI-based morphological classification of heterogeneous populations in breast cancer cells,” *Venture Research Program for Master's and PhD Students in the College of Engineering* (May 2021 – Dec. 2021) May 2021

Master's program research mentor — “The effect of 3D fibrous gelatin network property in activation of cancer-associated fibroblasts (CAFs) in breast cancer cells,” *Master's thesis, KAIST* Mar. 2019 – Feb. 2021

SKILLS

Experimental — Basic cell and spheroid culture, Live-cell microscopy, Collective cell micropatterning, Hydrogel fabrication, Cell traction force and monolayer stress microscopy, Atomic force microscopy (AFM), Soft-lithography fabrication, Fluorescent (confocal) microscopy, Light-sheet microscopy, Scanning electron microscopy (SEM), Electrospinning, Tensile stimulation apparatus development, Quantitative reverse-transcription PCR (RT-qPCR), Transfection

Computational — Image analysis with Python, Machine learning (deep learning, unsupervised clustering, supervised classification, dimensionality reduction) for time-series data, ImageJ, Illustrator, Jamovi, Origin, CAE software (ANSYS-CFX, ICEM-CFD, ANSYS-Mechanical, COMSOL Multiphysics)

Languages — Korean (native), English (professional working proficiency)

MILITARY SERVICE

Discharged as a sergeant, Republic of Korea Army · Incheon, Republic of Korea Mar. 2011 – Dec. 2012

REFERENCES

Prof. Hawa Racine Thiam (*Postdoctoral advisor*)

Departments of Bioengineering, Microbiology & Immunology, Stanford University

hrthiam@stanford.edu

Prof. Jennifer H. Shin (*Ph.D. program advisor*)

Department of Mechanical Engineering, KAIST

j_shin@kaist.ac.kr

Prof. Bomi Gweon (*Ph.D. dissertation committee member*)

Department of Mechanical Engineering, Sejong University

bgweon@sejong.ac.kr

Prof. Hyoung-Ho Kim (*B.S. and M.S. program mentor*)

School of Mechanical Material Convergence Engineering, Gyeongsang National University

khh106@gnu.ac.kr